# Disk Image Forensic Investigation using Autopsy

## Aim:

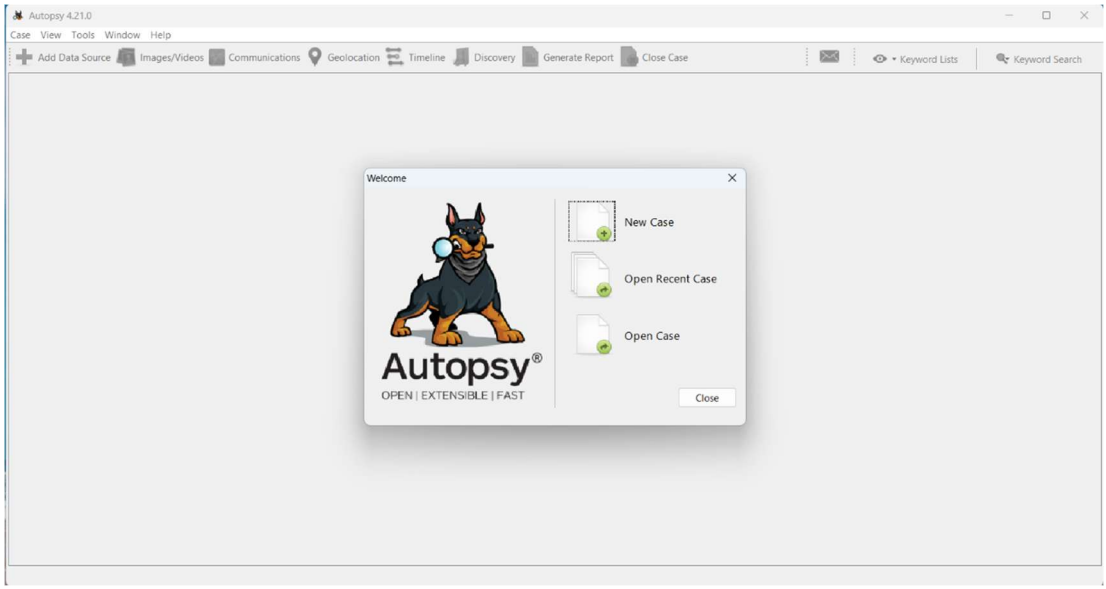
To conduct a digital forensics investigation using Autopsy focusing on file analysis, keyword searching, and timeline analysis.

## Requirements:

- System with Autopsy installed installed (Linux environment recommended).
- Disk image for analysis (obtained legally for educational or training purposes).

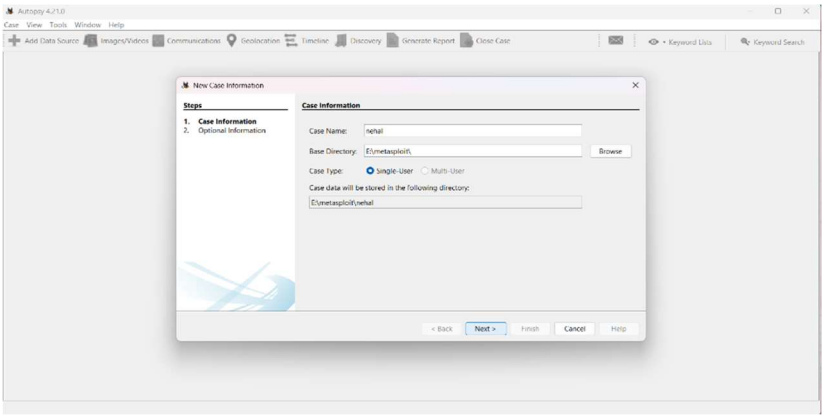
## Steps:

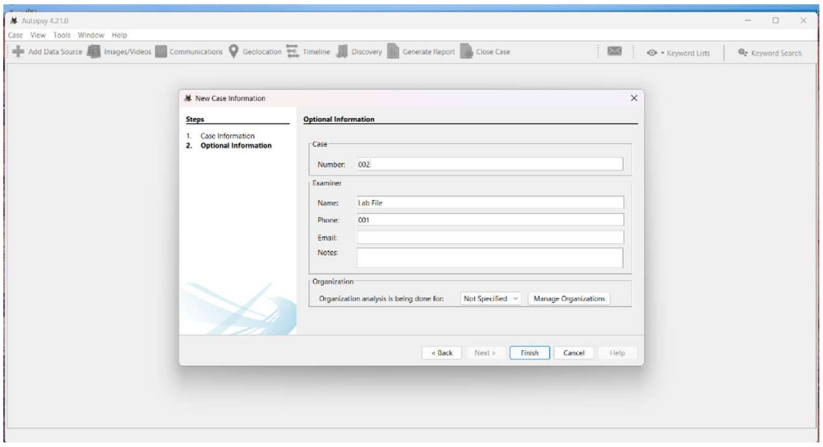
1. **Prepare the Lab Environment**: Install the Autopsy on a Linux system. Download and extract the disk image for analysis.




2. **Launch Autopsy** Open a terminal window and navigate to the directory where Autopsy is installed. Run the command "./autopsy" to start Autopsy.

3. **Create a New Case:** In the Autopsy interface, click on "New Case" and provide case details such as case name, examiner name, and case number. Select the location where the case files will be stored.


Autopsy 4.21.0
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: nehal
Base Directory: E:\metasploit\
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
E:\metasploit\nehal
< Back
Next >
Finish
Cancel
Help


Autopsy 4.21.0
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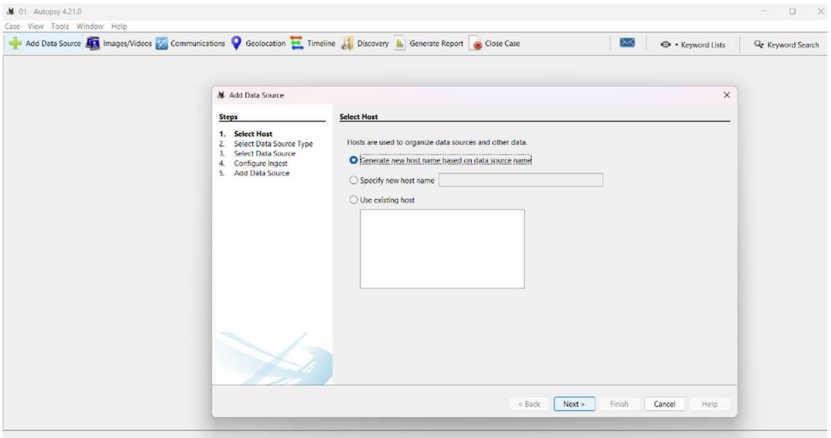
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number: 002
Examiner
Name: Lab File
Phone: 001
Email:
Notes:
Organization
Organization analysis is being done for:
Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help


01 - Autopsy 4.21.0
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
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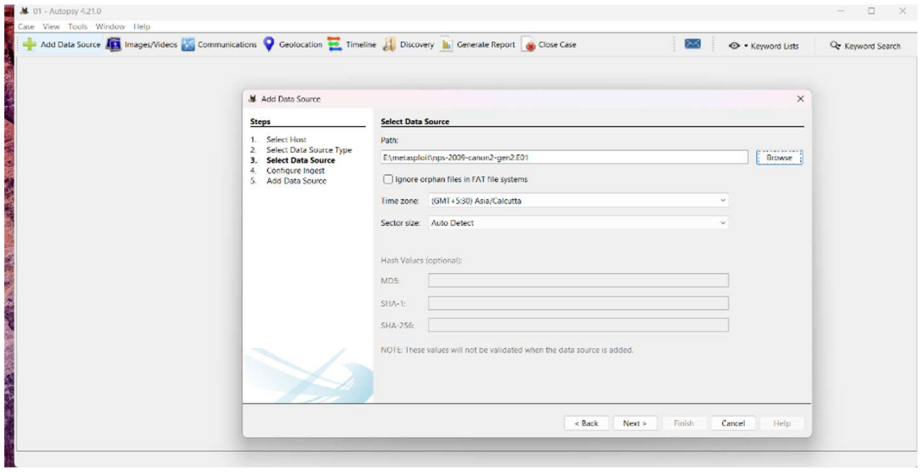
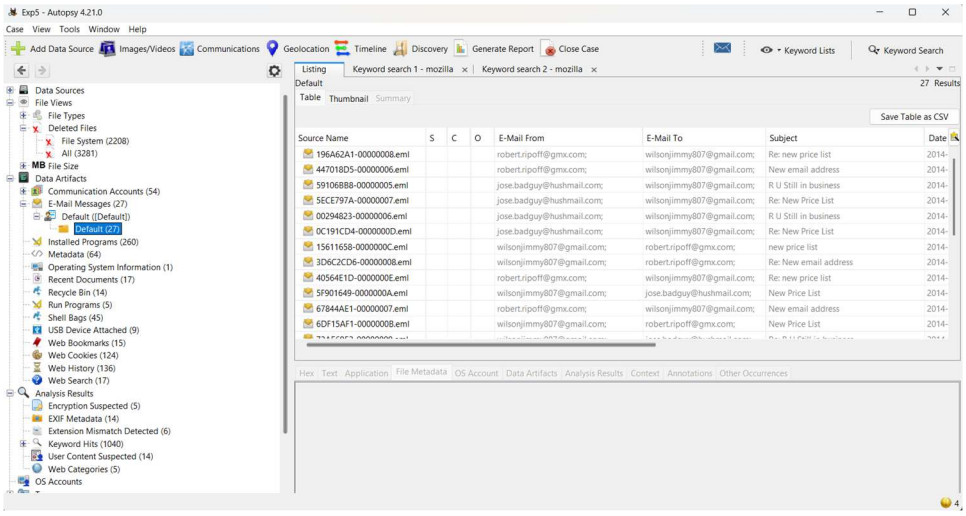
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help

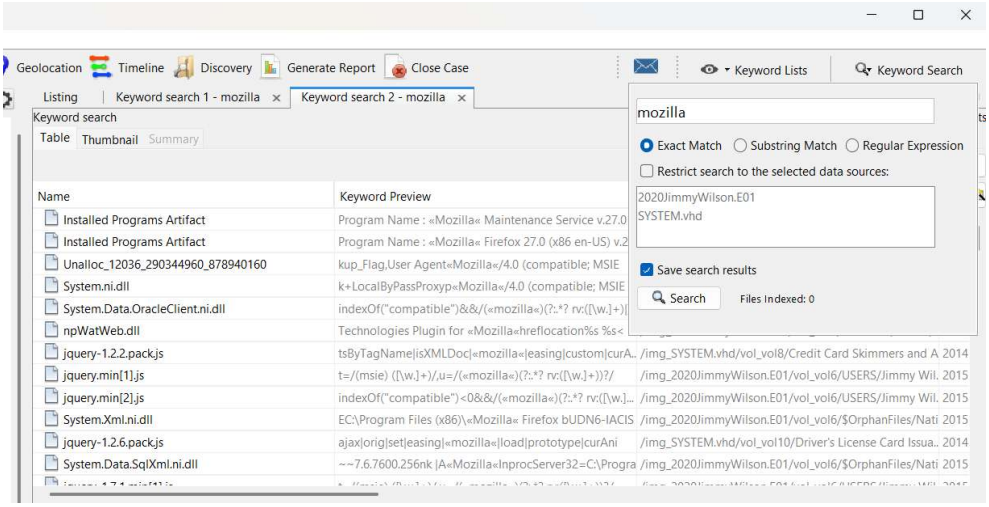



4. **Add Disk Image:** Click on “Add Host” and select “Disk Image” as the host type. Browse and select the disk image file for analysis. Autopsy will automatically create a data source for the disk image.

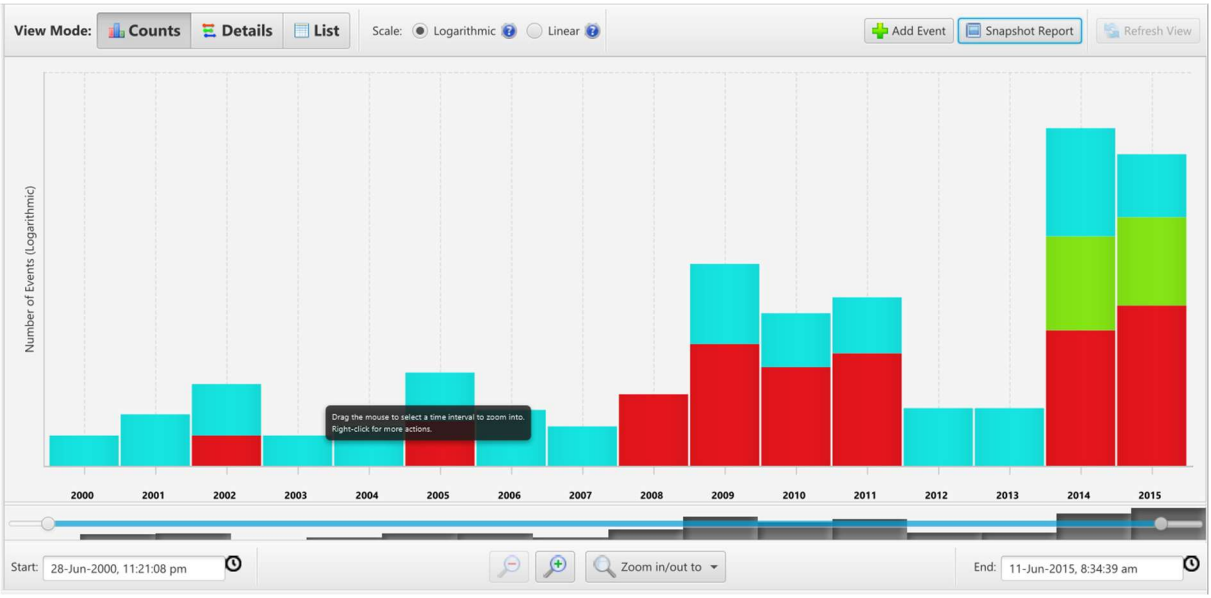



5. **Analyze File System:** Navigate to the “File Analysis” tab to explore the file system of the disk image. Use filters and search options to locate files of interest. Examine file attributes, such as file type, size, and timestamps.

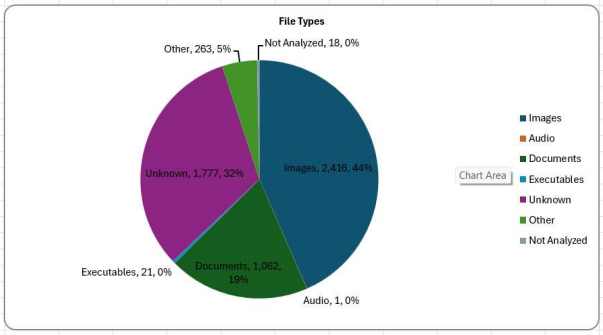
6. **Perform Keyword Search:** Switch to the “Keyword Search” tab and enter relevant keywords or phrases. Execute the search and review the search results. Analyze files containing the searched keywords for potential evidence.




7. **Generate Timeline:** Go to the “Timeline Analysis” tab and select the data source (disk image) for timeline generation. Configure timeline options such as time range and event types. Generate the timeline and analyze the chronological sequence of events.




8. **Generate Reports:** Navigate to the “Reports” tab and select the type of report to

generate (e.g., File Listing Report, Keyword Search Report, Timeline Report Customize report options and generate the reports. Review and save the reports for documentation.

| | |
|---|---|
| **Size** | 891.29 MB |
| | |
| **File Types** | **Count** |
| Images | 2,416 |
| Audio | 1 |
| Documents | 1,062 |
| Executables | 21 |
| Unknown | 1,777 |
| Other | 263 |
| Not Analyzed | 18 |
| | |
| **Allocated Files** | 4,336 |
| **Unallocated Files** | 1,222 |
| **Slack Files** | 2,994 |
| **Directories** | 5,128 |